

Project Design Phase-II Technology Stack (Architecture & Stack)

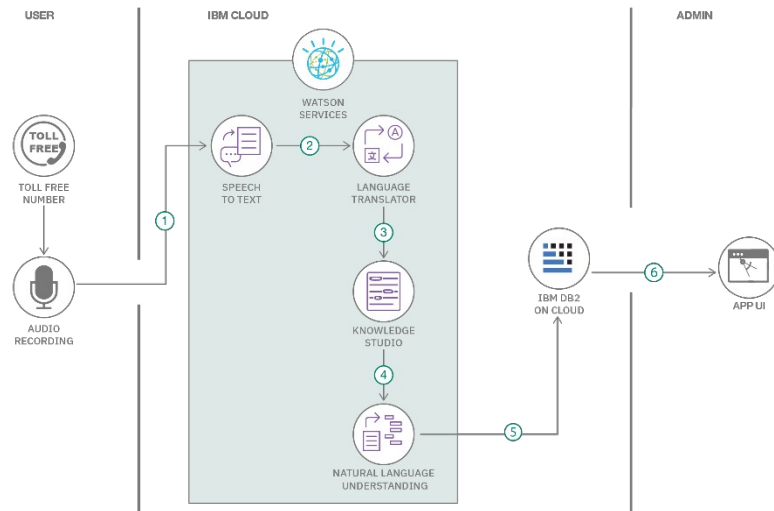
Date	30 September 2026
Team ID	SWTID-2026-5468
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

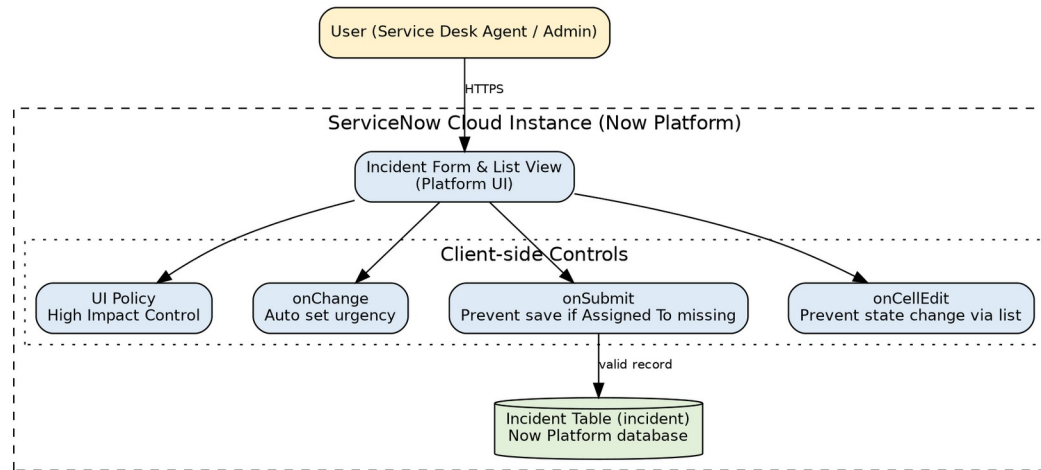
S.No	Component	Description	Technology
1.	User Interface	Incident form (Incident → Create New / Open) and Incident list view (Incident → All) where users enter, update and edit records	ServiceNow Platform UI (Now Platform)
2.	Application Logic-1	UI Policy "High Impact Control" (System UI → UI Policies): condition Impact is 1 - High; Assignment group mandatory; Reverse if false = true	ServiceNow UI Policy
3.	Application Logic-2	UI Policy Action on Urgency (Read-only = true) and onChange client script on Impact that sets Urgency to High and shows an info message	ServiceNow UI Policy Action; Client Script (JavaScript, g_form)
4.	Application Logic-3	onSubmit client script that blocks saving when Impact is High and Assigned To is empty, and onCellEdit client script that blocks State changes in the list	Client Scripts (JavaScript: onSubmit, onCellEdit)
5.	Database	Incident records with Impact, Urgency, Assigned To, Assignment group and State fields stored in the incident table	ServiceNow Now Platform database
6.	Cloud Database	Database is hosted and managed as part of the ServiceNow cloud instance; no separate database setup is needed	ServiceNow cloud instance
7.	File Storage	Not required for this project; no files are created or stored by the UI Policy or Client Scripts	Not applicable
8.	External API-1	Not used; all logic runs inside the ServiceNow instance	Not applicable

9.	External API-2	Not used	Not applicable
10.	Machine Learning Model	Not used; rules are fixed conditions on field values	Not applicable
11.	Infrastructure (Server / Cloud)	Configuration and testing done on a ServiceNow instance in the cloud, using an account with administrative or configuration access	ServiceNow cloud instance (Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source frameworks are used; the solution relies on native ServiceNow client-side features (UI Policies and Client Scripts) and the g_form API	ServiceNow UI Policies, Client Scripts
2.	Security Implementations	Rules are created and changed only with administrative or configuration access; incident access follows standard ServiceNow roles and access controls; save-time validation prevents incomplete high impact records	ServiceNow roles and access controls
3.	Scalable Architecture	Rules are defined once on the Incident table and apply to all records; they run in the user's browser, so no extra server load or setup as incidents grow	ServiceNow Now Platform, client-side scripting
4.	Availability	Controls are part of the instance configuration and are available whenever the Incident form and list are available; no external dependency	ServiceNow cloud instance
5.	Performance	Client-side checks respond immediately without a server round trip; scripts are short and run only for specific fields and events	Client Scripts, UI Policy

Project Technical Architecture:



Architecture Flow:

- The user opens the Incident form or list in the browser over HTTPS.
- The UI Policy evaluates Impact and applies field behaviour (Assignment group mandatory, Urgency read-only) and reverses it when the condition is no longer met.
- The onChange script sets Urgency to High and shows an info message when Impact changes to High.
- The onSubmit script checks Impact and Assigned To when the user saves, and stops the save with an error message if the data is incomplete.
- The onCellEdit script blocks State edits made from the list view.
- Only records that pass these checks are saved to the incident table.

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>